

SQL WINDOW FUNCTIONS PRACTICE

DATASET

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    department_id INT,  
    salary INT  
);
```

```
INSERT INTO employees VALUES  
(1, 'Amit', 10, 50000),  
(2, 'Ravi', 10, 60000),  
(3, 'Neha', 20, 55000),  
(4, 'Priya', 20, 70000),  
(5, 'Arjun', 30, 65000),  
(6, 'Meera', 30, 62000),  
(7, 'Kunal', 10, 58000),  
(8, 'Divya', 20, 72000);
```

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    amount INT  
);
```

```
INSERT INTO orders VALUES  
(101, 1, '2026-06-01', 2000),  
(102, 2, '2026-06-02', 1500),  
(103, 1, '2026-06-03', 3000),  
(104, 3, '2026-06-04', 2500),  
(105, 2, '2026-06-05', 1800),  
(106, 3, '2026-06-06', 2200),  
(107, 1, '2026-06-07', 2700),  
(108, 2, '2026-06-08', 1900);
```

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(50),  
    class_id INT,  
    exam_date DATE,  
    marks INT  
);
```

```
INSERT INTO students VALUES  
(1, 'Rahul', 101, '2026-06-01', 78),  
(2, 'Sneha', 101, '2026-06-02', 85),  
(3, 'Karan', 102, '2026-06-01', 90),  
(4, 'Pooja', 102, '2026-06-03', 88),  
(5, 'Vikas', 103, '2026-06-02', 76),  
(6, 'Anita', 103, '2026-06-04', 82),  
(7, 'Manish', 101, '2026-06-05', 92),  
(8, 'Ritu', 102, '2026-06-06', 95);
```

```
CREATE TABLE products (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(50),  
    category_id INT,  
    sales INT  
);
```

```
INSERT INTO products VALUES  
(1, 'Laptop', 1, 50000),  
(2, 'Tablet', 1, 30000),  
(3, 'Phone', 2, 40000),  
(4, 'Headphones', 2, 15000),  
(5, 'Camera', 3, 35000),  
(6, 'Smartwatch', 3, 20000);
```

QUESTIONS

Beginner (1-7)

1. Show each employee's salary along with the average salary of their department.
2. For each order, calculate the cumulative sales amount up to that date.
3. List students with their marks and the highest marks in their class.

4. Display each product's sales and the total sales of its category.
5. Rank employees within their department by salary.
6. For each order, calculate the cumulative average order amount over time.
7. Assign a row number to each student within their class based on marks.

Intermediate (8-14)

8. Find employees whose salary is above or below their department average.
9. For each order, compute the moving average of the last 3 orders.
10. Show each product's sales and its percentage contribution to category sales.
11. Rank employees across the company by salary, ensuring ties get the same rank.
12. For each order, calculate the difference compared to the previous order's amount.
13. For each order, show the next order's amount.
14. Display students' marks along with the highest marks achieved so far in exam sequence.

Advanced (15-20)

15. Calculate the percentile rank of each employee's salary across the company.
16. Show each product's sales and its cumulative distribution within category.
17. For each order, compute the total sales in the last 7 days relative to order_date.
18. For each student, calculate the gap between their marks and the class topper's marks.
19. Find the 3rd highest salary in each department and display it alongside employees.
20. For each order, calculate the ratio of its amount to total company sales.